Chapter 7

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7.1. Let A be a non-Noetherian Ring and let Σ be the set of ideals in A which are not finitely generated. Show that Σ has maximal elements, and that the maximal elements of Σ are prime ideals. Hence, a ring in which every prime ideal is finitely generated is Noetherian (I. S. Cohen).

Proof. We begin by showing Σ has maximal elements. It is nonempty by (6.2), so it suffices to show that each chain $\{I_n\} \subset \Sigma$ has an upper bound in Σ. Namely, we must show that the ideal $I = \bigcup_n I_n$ is not finitely generated. Suppose not, and take generators $x_1, \ldots, x_m \in I$ Each $x_i \in I_{n_i}$, so we have $N \geqslant 1$ such that $x_i \in I_N$ for all $i = 1, \ldots, m$, whence

$$I_N \subset I = (x_1, \ldots, x_m) \subset I_N$$

contradicts the assumption that I_N is not finitely generated. Zorn's lemma supplies us with maximal elements of Σ.

Take a maximal element $I \in \Sigma$, and suppose $xy \in I$ for $x, y \notin I$. Consider the ideals $(I, x) \supsetneq I$ and $(I, y) \supsetneq I$. Maximality of I implies both are finitely generated, so that $(I, x)(I, y) = (I^2, xI, yI, xy) \subset I$ is finitely generated. □

7.3 Let $\mathfrak{a}$ be an irreducible ideal in a ring A. Then the following are equivalent:
- (a) $\mathfrak{a}$ is primary;
- (b) for every multiplicatively closed subset S of A, we have $(S^{-1}\mathfrak{a})^c = (\mathfrak{a} : x)$ for some $x \in S$;
- (c) the sequence $(\mathfrak{a} : x^n)$ is stationary, for every $x \in A$.

7.8 If $A[x]$ is Noetherian, is A necessarily Noetherian?

7.9 Let A be a ring such that
- (a) for each maximal ideal $\mathfrak{m}$ of A, the local ring $A_{\mathfrak{m}}$ is Noetherian;
- (b) for each $x \neq 0$ in A, the set of maximal ideals of A which contain x is finite.

Show that A is Noetherian.

Proof. □

7.12 Let A be a ring and B a faithfully flat A-algebra (Ex. 3.16). If B is Noetherian, show that A is Noetherian.

Proof. Let $f : A \to B$ be a faithfully flat Noetherian A-algebra, and take an ideal $\mathfrak{a}$ of A. (6.2) gives us that $\mathfrak{a}^e \subset B$ is finitely generated as a B-module, and B faithfully flat implies $\mathfrak{a}^{ec} = \mathfrak{a}$.

Take a minimal set of generators $(b_1 a_1, \ldots, b_n a_n) = \mathfrak{a}^e$ with $a_i \in \mathfrak{a}$ and $b_i \in B$. For each $j = 1, \ldots, n$, we have

$$(1 - b_j' b_j) a_j = \sum_{i \neq j} b_i' b_i a_i,$$

so $a_j = b_j' b_j a_j$ by minimality. Then the $a_1, \ldots, a_n \in A \subset B$ generate $\mathfrak{a}^e$ as a B-module, and hence generate $\mathfrak{a}^{ec} = \mathfrak{a}$ as an A-module. We conclude that any submodule $\mathfrak{a}$ of A is finitely generated, whence A itself is Noetherian. □

7.13 Let $f : A \to B$ be a ring homomorphism of finite type and let $f^* : \operatorname{Spec} B \to \operatorname{Spec} A$ be the mapping associated with f. Show that the fibers of f^* are Noetherian subspaces of $\operatorname{Spec} B$.

7.15. Let A be a Noetherian local ring, $\mathfrak{m}$ its maximal ideal, and k its residue field, and let M be a finitely generated A-module. Then the following are equivalent:
(i) M is free;
(ii) M is flat;
(iii) the mapping $\mathfrak{m} \otimes M \to A \otimes M$ is injective;
(iv) $\operatorname{Tor}_1^A(k, M) = 0$.

Proof. (i) $\implies$ (ii): A is flat, and a direct sum of flat A-modules is flat.
(ii) $\implies$ (iii): This is clear from the definition of flat.
(iii) $\implies$ (iv): Exactness of

$$0 \to \mathfrak{m} \to A \to k \to 0$$

gives us exactness of

$$\operatorname{Tor}_1^A(k, M) \to \mathfrak{m} \otimes M \to A \otimes M \to k \otimes M \to 0.$$

The first map is injective by assumption, so in fact $\operatorname{Tor}_1^A(k, M) = 0$, as desired.
(iv) $\implies$ (i): A Noetherian gives us M finitely generated, so we can take a minimal set of generators $x_1, \ldots, x_n \in M$ of M. Then (iv) allows us to apply $k \otimes_A (\cdot)$ to the short exact sequence $0 \to \ker \phi \to A^n \to M \to 0$ and receive a short exact sequence

$$0 \to \frac{\ker \phi}{\mathfrak{m}\ker \phi} \to k^n \to \frac{M}{\mathfrak{m}M} \to 0.$$

Any set of generators of $M/\mathfrak{m}M$ lifts to a set of generators for M by (2.8). Then minimality of $\{x_i\}_{i=1}^n \subset M$ gives us minimality of the $\{\overline{x}_i\}_{i=1}^n \subset M/\mathfrak{m}M$. Hence, $\{\overline{x}_i\}_{i=1}^m$ form a basis of $M/\mathfrak{m}M$, so that

$$0 \cong \ker\left(k^n \to M/\mathfrak{m}M\right) \cong \frac{\ker \phi}{\mathfrak{m}\ker \phi}.$$

It follows by Nakayama's lemma that $\ker \phi = 0$, whence $\phi : A^n \to M$ is an isomorphism. ☐

7.16. Let A be a Noetherian ring, and let M be a finitely generated A-module. Then the following are equivalent:
(i) M is a flat A-mdoule;
(ii) $M_{\mathfrak{p}}$ is a free $A_{\mathfrak{p}}$-module for all $\mathfrak{p} \in \operatorname{Spec} A$;
(iii) $M_{\mathfrak{m}}$ is a free $A_{\mathfrak{m}}$-module, for all maximal ideal $\mathfrak{m}$.
In other words, flat = locally free.

Proof. This is a corollary of (3.10) and (Ex. 7.15). ☐

7.20 Let X be a topological space and let $\mathscr{F}$ be the smallest collection of subsets of X which contain all open subsets of X and is closed with respect to the formation of finite intersections and complements.
(a) Show that a subset E of X belongs to $\mathscr{F}$ if and only if E is a finite union of sets of the form $U \cap C$, where U is open and C is closed.
(b) Suppose that X is irreducible and let $E \in \mathscr{F}$. Show that E is dense in X if and only if E contains a non-empty open set in X.

Proof. (a) Let $E \in \mathscr{F}$. Then we can write

$$E = \left(\bigcap_{i \in I} \left(\bigcap_{j \in J_i} C_k \right) \right)^c \cap \bigcap_{i=1}^{n_1} ()$$

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(b) Clearly any such $E \in \mathscr{F}$ is dense, as any nonempty open of X is dense. Conversely, take $E = \bigcup_{i=1}^n U_i \cap C_i$ such that $\overline{E} = X$. □

7.21 Let X be a Noetherian topological space and let $E \subset X$. Show that $E \subset \mathscr{F}$ if and only if, for each irreducible closed set $X_0 \subset X$, either $\overline{E \cap X_0} \neq X_0$ or else $E \cap X_0$ contains a non-empty open subset of X_0. The sets belonging to $\mathscr{F}$ are called the *constructible subsets* of X.

7.22 Let X be a Noetherian topological space and let E be a subset of X. Show that E is open in X if and only if, for each irreducible closed subset X_0 in X, either $E \cap X_0 = \emptyset$ or $E \cap X_0$ contains a non-empty open subset of X_0.

7.23 Let A be a Noetherian ring, $f : A \to B$ a ring homomorphism of finite type (so that B is Noetherian). Let $X = \operatorname{Spec} A$, $Y = \operatorname{Spec} B$, and let $f^* : Y \to X$ be the mapping associated with f. Then the image under f^* of a constructible subset E of Y is a constructible subset of X.

7.24 With the notation and hypotheses of (Ex. 7.23), f^* is an open mapping if and only if f has the going down property.

7.25 Let A be Noetherian, $f : A \to B$ of finite type and flat. Then $f^* : \operatorname{Spec} B \to \operatorname{Spec} A$ is an open mapping.

Proof. By (Ex. 5.11), f has the going down property, whence f^* is an open mapping by (Ex. 7.24). □